Transcriptional noise in gene expression can induce rare phenotypic switching events in heterogenous cancer cells with epithelial-mesenchymal heterogeneity

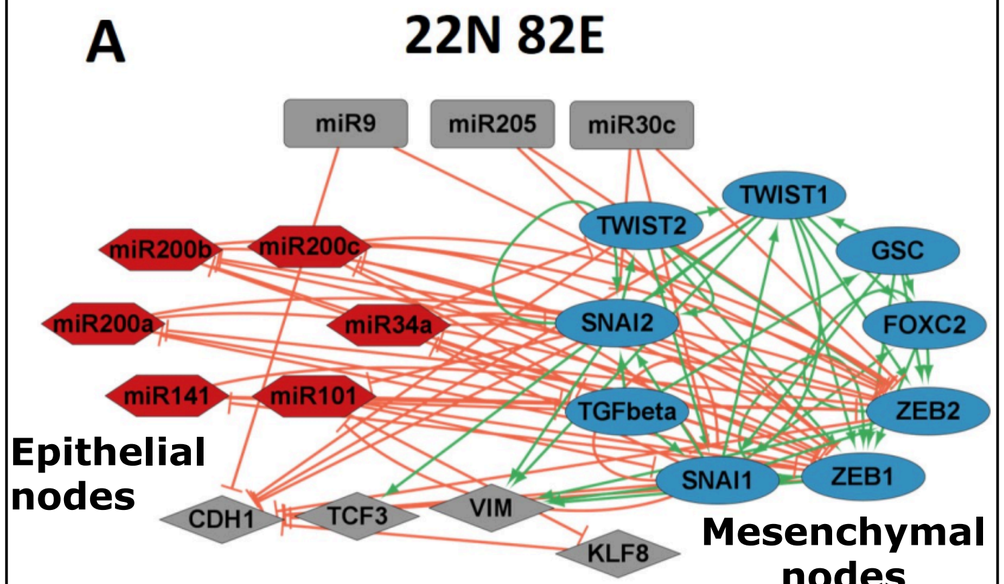
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Supervisors:
Paras Jain, Post Doc,
Prof. Mohit Kumar Jolly, Head of
Cancer Systems Biology lab



BACKGROUND & MOTIVATION

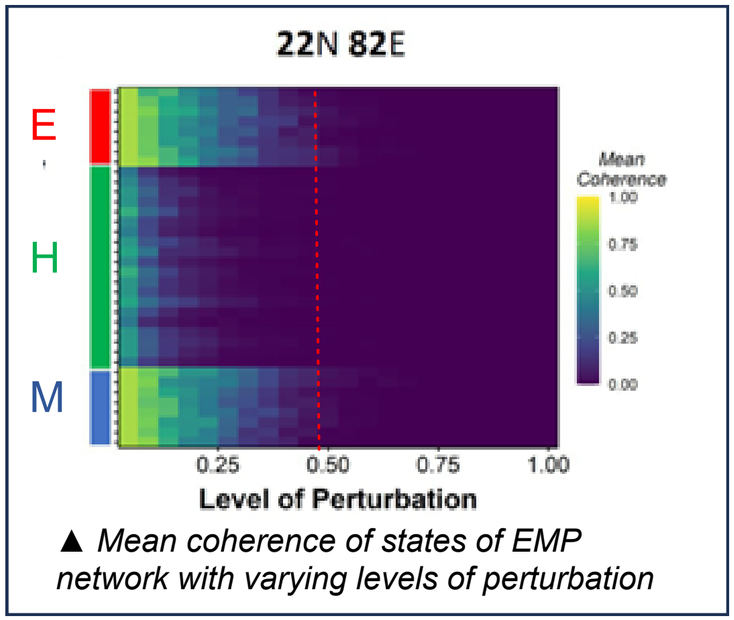
GENE REGULATORY NETWORK exhibiting TEAMS structure (involved in Cancer metastasis)



▲ *EMP network with 22 Nodes and 82 Edges*

Boolean simulations reveal that

>> TEAMS network is highly ROBUST
>> TERMINAL states are highly stable
>> Resistant to PERTURBATIONS



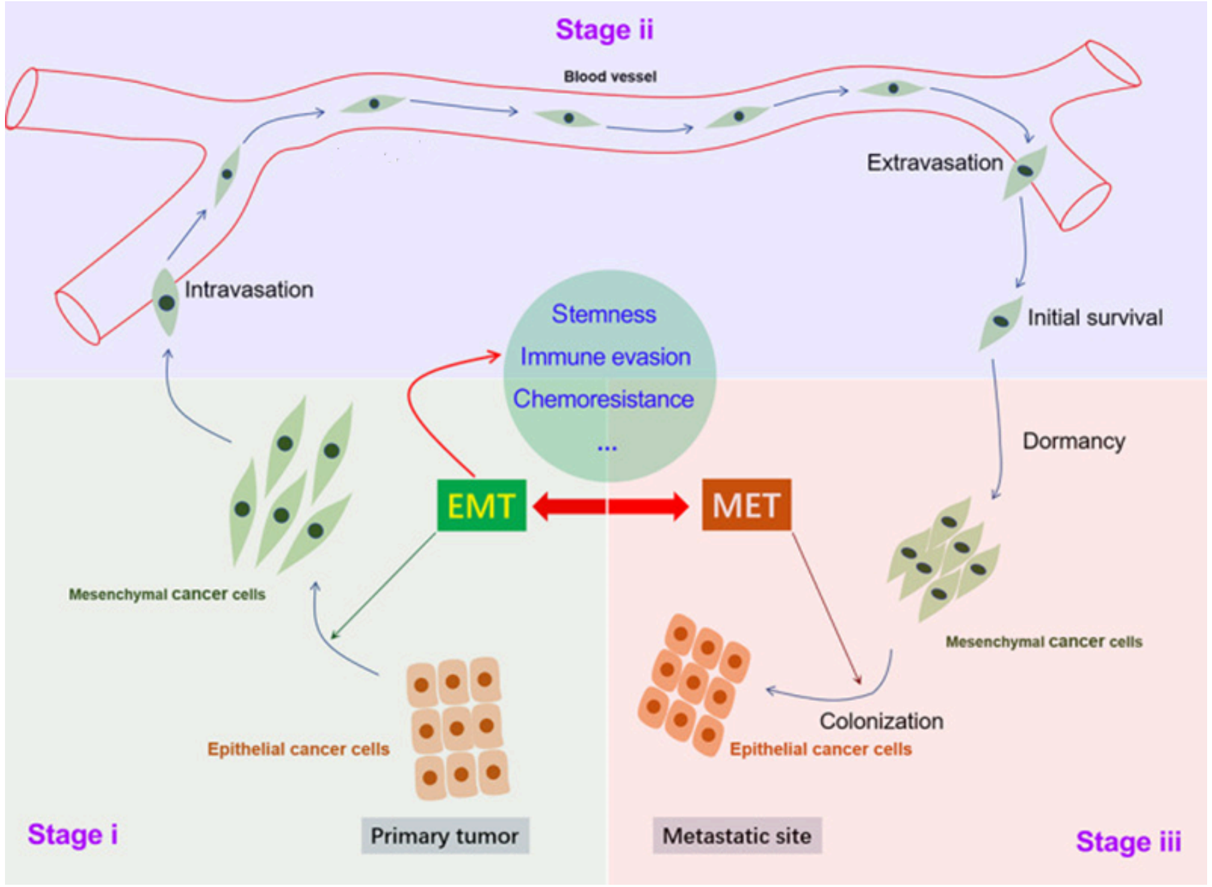
▲ *Mean coherence of states of EMP network with varying levels of perturbation*

Large amount of external perturbation needed to destabilize the TEAMS structure

Landscape of epithelial–mesenchymal plasticity as an emergent property of coordinated teams in regulatory networks

Kishore Hari[1], Varun Ullanat[2], Archana Balasubramanian[3], Aditi Gopalan[2], Mohit Kumar Jolly[1]*

Cancer cells metastasize and invade other organs



Cancer deaths in Norway 2015

	All cancer deaths		Cancer deaths with metastases		
	Male	Female	Male (%)	Female (%)	Total (%)
All cancer	5810	4936	3390 (58.3)	3118 (63.2)	6508 (60.1)
Solid tumors[a]	5229	4493	3374 (64.5)	3109 (69.2)	6483 (66.7)
Colon	536	600	445 (83.0)	466 (77.7)	911 (80.2)
Lung/trachea	1169	973	918 (78.5)	747 (76.8)	1665 (77.7)
Breast	6	583	5 (83.3)	440 (75.5)	445 (75.6)
Ovary	0	282	0	255 (90.4)	255 (90.4)
Prostate	1034	0	519 (50.2)	0	519 (50.2)
CNS	199	165	25 (12.6)	9 (5.5)	34 (9.3)

Despite the inherent robustness of the TEAMS structure, what drives the spontaneous switching of cancer cells between terminal and hybrid phenotypic states?

Heterogeneity and plasticity of epithelial–mesenchymal transition (EMT) in cancer metastasis: Focusing on partial EMT and regulatory mechanisms

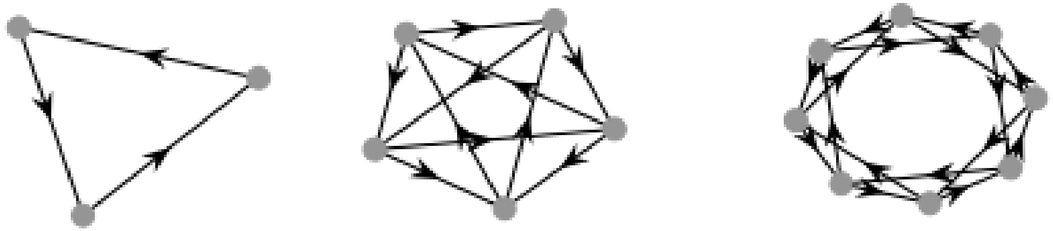
Dandan Li [1,2], Lingyun Xia [1], Pan Huang [2], Zidi Wang [2], Qiwei Guo [2], Congcong Huang [2], Weidong Leng [1,✉], Shanshan Qin [1,2,✉]

HYPOTHESIS

1. **CONNECTIVITY and SIZE of the underlying gene regulatory network**
2. **Influence of TRANSCRIPTIONAL NOISE**

account for the emergence of **epithelial–mesenchymal heterogeneity** within a cancer cell population.

SIZE = No. of nodes
CONNECTIVITY = No. of incoming edges



Transcriptional Noise
can be modelled in different ways

>> Two-State (**ON–OFF**) Promoter Model
>> **Intrinsic Noise (Chemical Langevin Equation)**
>> Extrinsic Noise (stochastic degradation rate)
>> Explicit noise term (Stochastic differential equation)

In this work, we will consider:

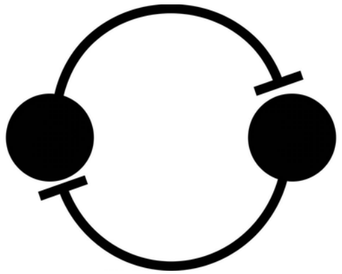
- **Toggle switch**
- **Intrinsic noise (CLE)**

APPROACH

COMPUTATIONAL PIPELINE → PARAMETER GENERATION → NUMERICAL SIMULATIONS → DATA ANALYSIS → QUANTIFICATION THROUGH STATISTICAL METRICS

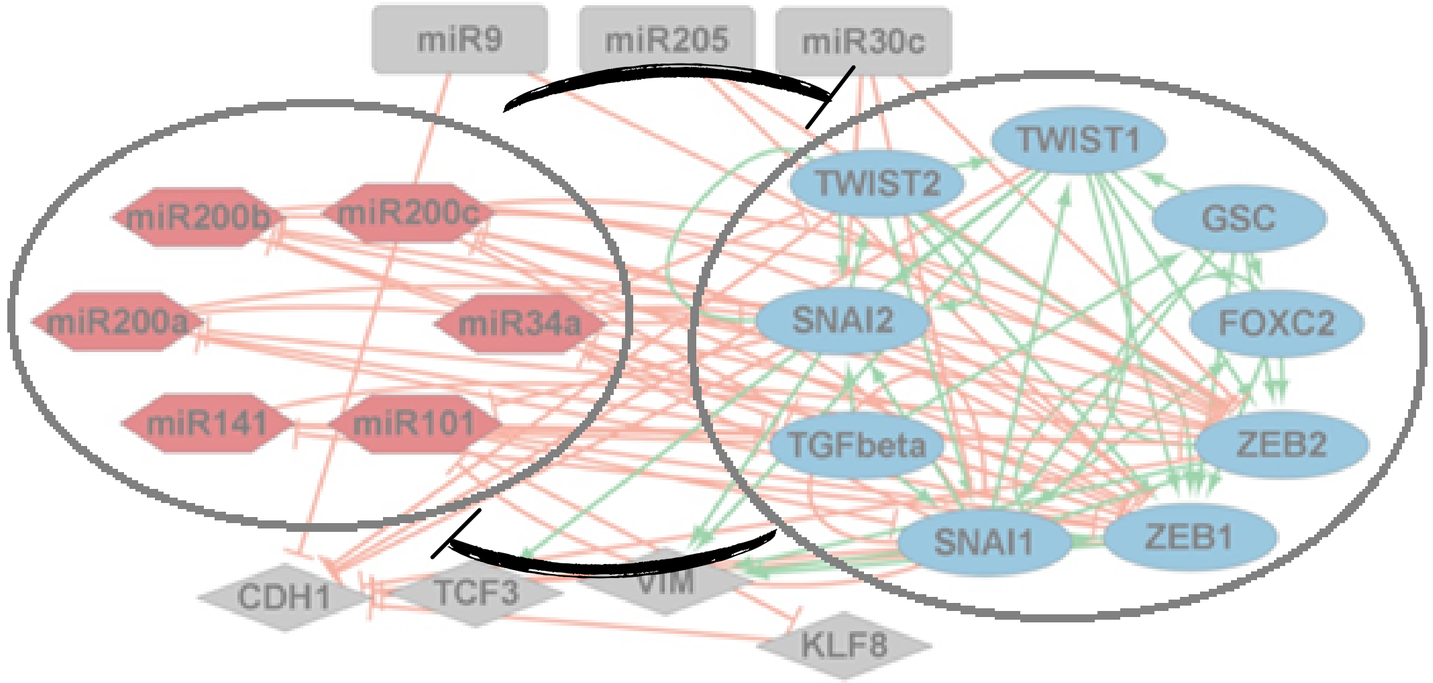
METHODS

NETWORK MOTIF BEING ANALYZED



SIMPLIFIED

Toggle Switch Network Motif



TEAMS Network Motif

SIMULATION: Stochastic Differential Equations

Ordinary Differential Equations

$$\frac{dx(t)}{dt} = f(x(t), t)$$

- Describes the **deterministic** evolution of a system over time.
- Same initial conditions always leads to the same outcome

Stochastic Differential Equations

$$dx(t) = \underbrace{f(x(t), t)\,dt}_{\text{DRIFT term}} + \underbrace{g(x(t), t)\,dW(t)}_{\text{DIFFUSION coefficient}}$$

Wiener process

- Captures random fluctuations or “noise” in the system
- We consider that strength of noise depends on the current state of the system

Chemical Langevin Equation [CLE]

Process: mRNA transcription and degradation

$$\frac{dm(t)}{dt} = \{birth_process + birth_noise\} - \{death_process + death_noise\}$$

PROPERTIES of CLE

- Moderate/high copy number of molecules
- Gaussian noise terms
- Captures **intrinsic noise** of system
- Solved using **Euler – Maruyama scheme**
- Computationally faster than exact algorithms like **Gillespie's SSA**,

Defining the set of SDE's for simulation

production — degradation

$$\frac{dm_1(t)}{dt} = \left\{ g \cdot \left(1 - \left(1-\lambda\right) \cdot H\left(m_2\right)\right) + \sigma_{birth,1} \cdot N\left(0,1\right) \right\} - \left\{ k \cdot m_1 + \sigma_{death,1} \cdot N\left(0,1\right) \right\}$$

$$\frac{dm_2(t)}{dt} = \left\{ g \cdot \left(1 - \left(1-\lambda\right) \cdot H\left(m_1\right)\right) + \sigma_{birth,2} \cdot N\left(0,1\right) \right\} - \left\{ k \cdot m_2 + \sigma_{death,2} \cdot N\left(0,1\right) \right\}$$

$$\sigma_{birth,i} = \sqrt{g \cdot \left(1 - \left(1-\lambda\right) \cdot H\left(m_j\right)\right)}$$

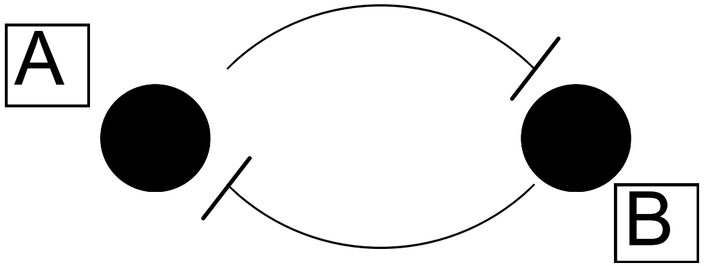
$$\sigma_{death,i} = \sqrt{k \cdot m_i}$$

Std. deviation of noise

$$H(m) = \frac{m^n}{m^n + K^n}$$

Hill's function

Shifted Hill regulation



Toggle Switch Network Motif

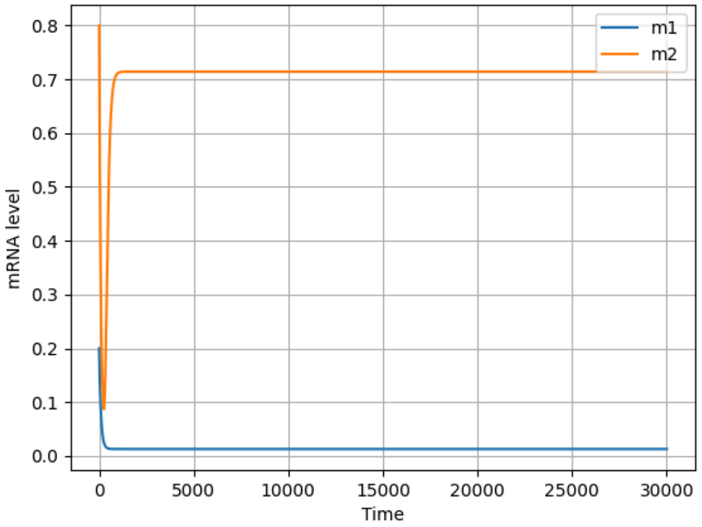
Initial Conditions

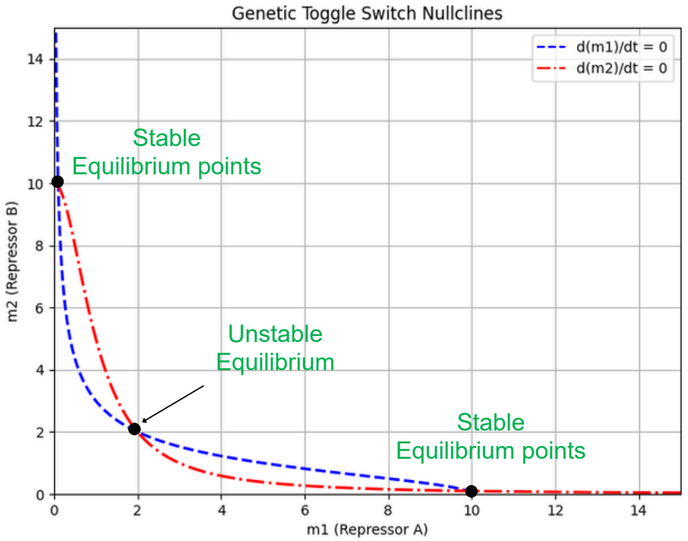
>> Simulation starts from the **steady state**

>> Steady states obtained from ODE simulation

$$\frac{dm_1(t)}{dt} = g \cdot \left(regulation\right) - k \cdot m_1$$

$$\frac{dm_2(t)}{dt} = g \cdot \left(regulation\right) - k \cdot m_2$$





Parameter sets

Parameter sets are generated using **Half-functional rule**

>> Behaviour of network is susceptible to parameter sets
>> Particularly for $\mathbf{K_1}$, $\mathbf{K_2}$ values
>> Parameters are chosen such that **regulatory interactions** are functional about **HALF** the time across parameter sets

g : production rate
k : degradation rate
n : Hill's coefficient
λ^+ : Fold change (activating links)
λ^- : Fold change (inhibiting links)
K_1 : threshold of link $2 \rightarrow 1$
K_2 : threshold of link $1 \rightarrow 2$

para_indx	g	k	n	lambda_p	lambda_n	thrs_1	thrs_2
1	37.56698	0.012717	1	63.55764	0.010921	40.80127	72.70262
2	8.826715	0.092736	2	79.58027	0.053785	54.51301	49.51901
3	64.48101	0.012463	2	15.96025	0.051572	62.82468	50.84236
4	30.31592	0.015113	6	15.34401	0.044228	70.45334	72.40897
5	42.81008	0.015315	3	81.48879	0.024344	6.004771	86.20897
6	22.91491	0.012226	6	55.32122	0.017769	58.38122	66.42193
7	38.35198	0.013742	2	59.44031	0.059576	62.0054	67.83281
8	84.10922	0.022018	3	81.33049	0.030089	62.70054	63.23859
9	62.24478	0.01266	2	54.84606	0.025954	85.60562	31.09755
10	70.31049	0.01545	5	63.36704	0.020467	13.13168	70.2508
11	1.032747	0.018363	1	3.962626	0.010403	62.55905	61.40599
12	51.91626	0.010086	5	12.45579	0.010989	33.45121	24.13375
994	74.54036	0.01721	3	31.66834	0.011762	14.57284	43.60159
995	71.44862	0.08778	3	87.96433	0.018438	79.67644	68.36272
996	72.02874	0.0196	2	50.4201	0.034416	70.65186	81.57008
997	53.95407	0.050277	3	2.484309	0.012127	68.02396	30.24666
998	79.45262	0.013369	4	38.63676	0.019201	70.08309	12.12944
999	99.45778	0.013252	3	11.83259	0.011489	81.17749	54.20636
1000	76.21721	0.04617	3	60.80714	0.052836	42.51629	73.32474

para indx	x_1	x_2	type
4	89	526	1
4	473	89	2
6	33	1846	1
6	1813	33	2
8	119	596	1
8	580	120	2
21	5	371	1
21	387	5	2
22	104	1669	1
22	3813	104	2
23	22	730	1
23	721	22	2
26	25	750	1
26	862	25	2
27	5	101	1
27	96	19	2
28	18	864	1
28	813	18	2
979	52	2190	1
979	1178	52	2
982	8	652	1
982	651	8	2
987	31	917	1
987	1079	31	2
991	51	2617	1
991	2579	51	2
992	49	4276	1
992	3932	49	2
995	16	804	1
995	808	15	2
997	13	989	1
997	1066	13	2
999	87	1517	1
999	3459	86	2

Simulations are conducted with . . .

Noise added to ONE node

$$\frac{dm_1(t)}{dt} = \left\{g \cdot \left(1-(1-\lambda)\cdot H(m_2)\right) + \sigma_{birth,1}\cdot N(0,1)\right\} - \left\{k\cdot m_1 + \sigma_{death,1}\cdot N(0,1)\right\}$$

$$\frac{dm_2(t)}{dt} = \left\{g \cdot \left(1-(1-\lambda)\cdot H(m_1)\right)\right\} - \left\{k\cdot m_2\right\}$$

Noise added to BOTH nodes

$$\frac{dm_1(t)}{dt} = \left\{g \cdot \left(1-(1-\lambda)\cdot H(m_2)\right) + \sigma_{birth,1}\cdot N(0,1)\right\} - \left\{k\cdot m_1 + \sigma_{death,1}\cdot N(0,1)\right\}$$

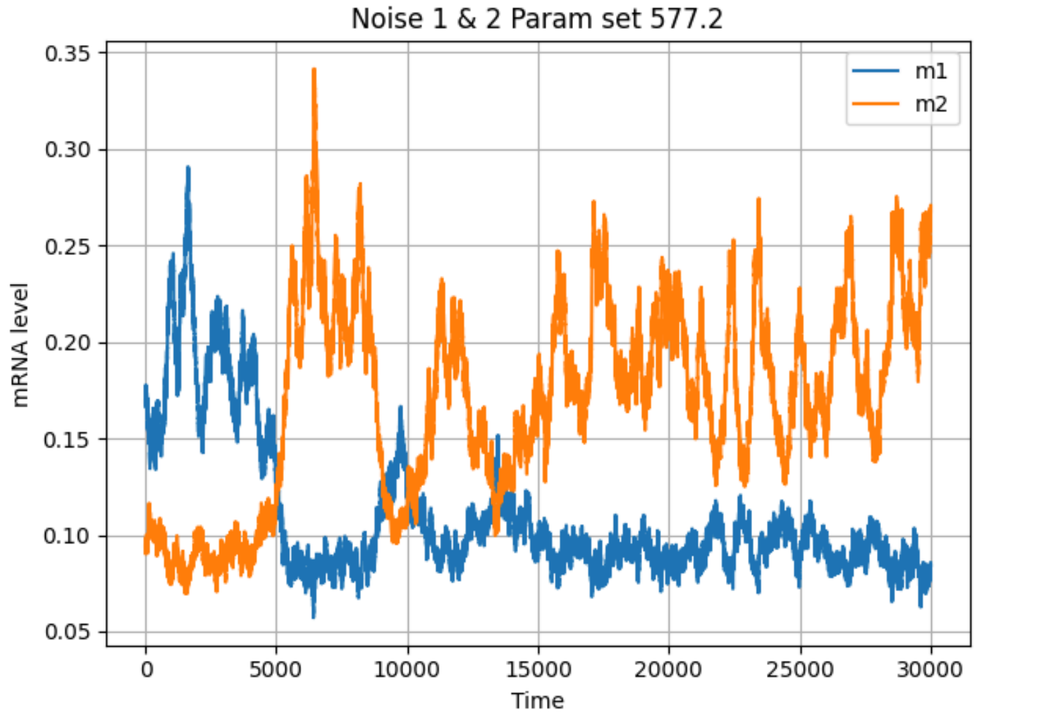
$$\frac{dm_2(t)}{dt} = \left\{g \cdot \left(1-(1-\lambda)\cdot H(m_1)\right) + \sigma_{birth,2}\cdot N(0,1)\right\} - \left\{k\cdot m_2 + \sigma_{death,2}\cdot N(0,1)\right\}$$

Settings of the simulation

Total simulation time: **30,000** time points

Time step: $\Delta t = \dfrac{1}{\text{production rate}}$

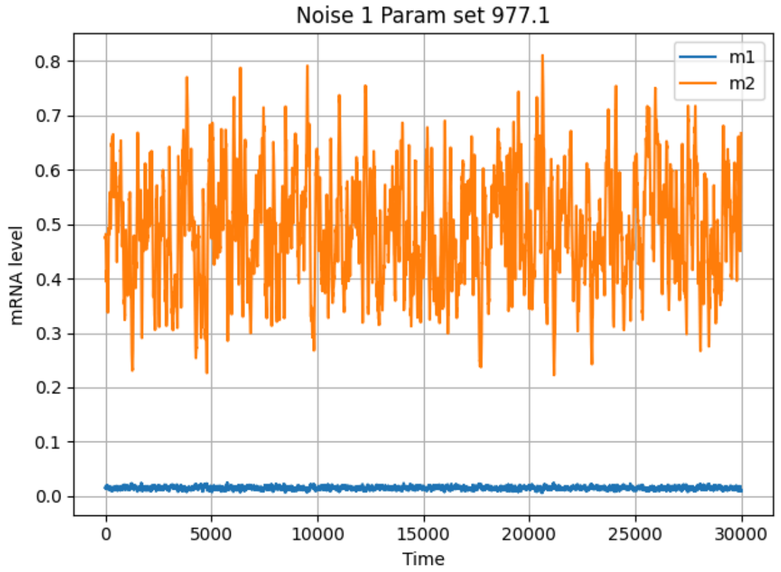
>> Simulations were conducted over 700 parameter sets
>> Parallelized over 12 CPU cores to improve time complexity
>> 30000 points of data saved for analysis to improve memory usage



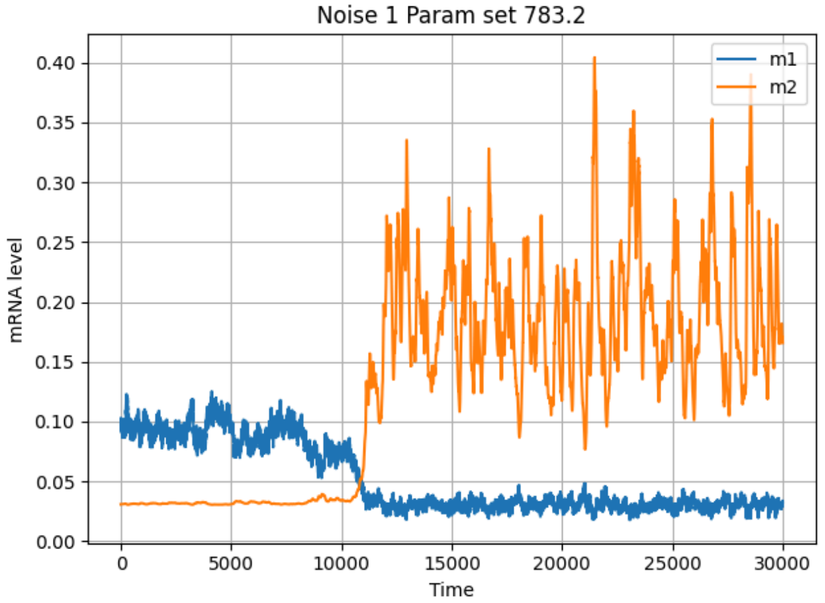
$$\text{mRNA levels normalized over max copy number} = \frac{\text{production rate}}{\text{degradation rate}}$$

TYPES OF EXPRESSION PATTERNS

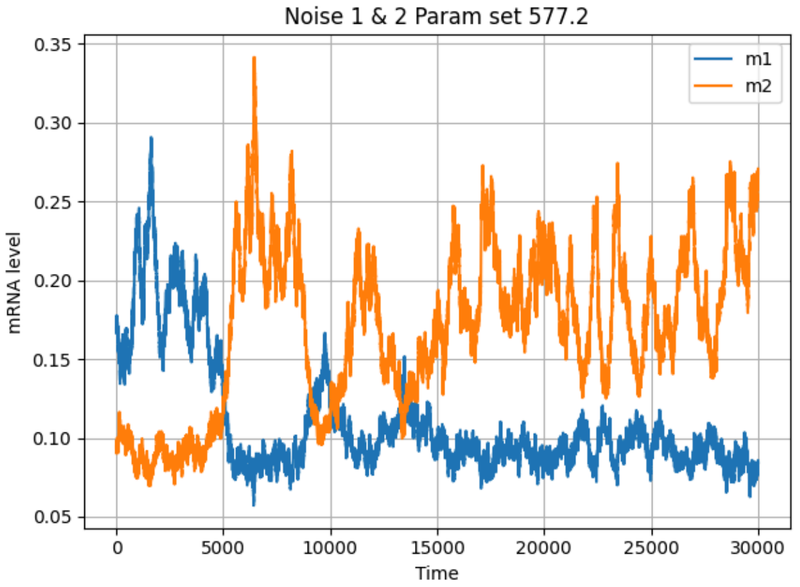
Terminal states maintained

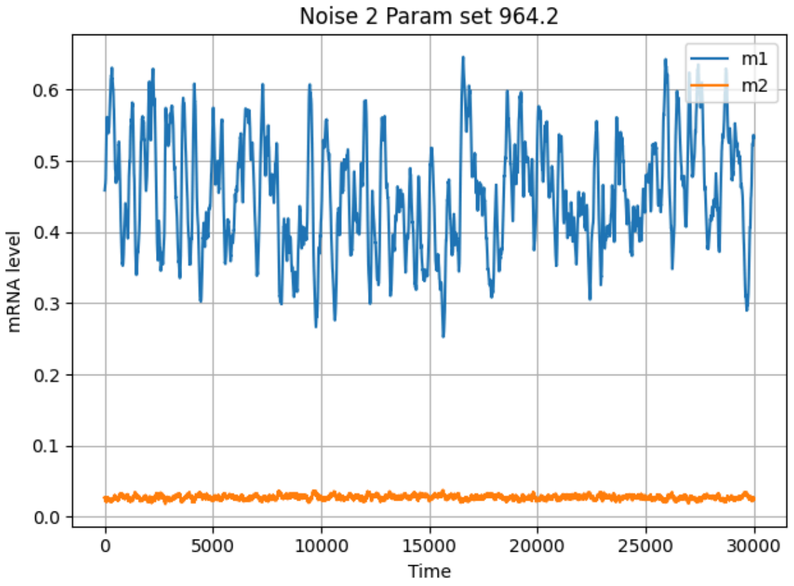


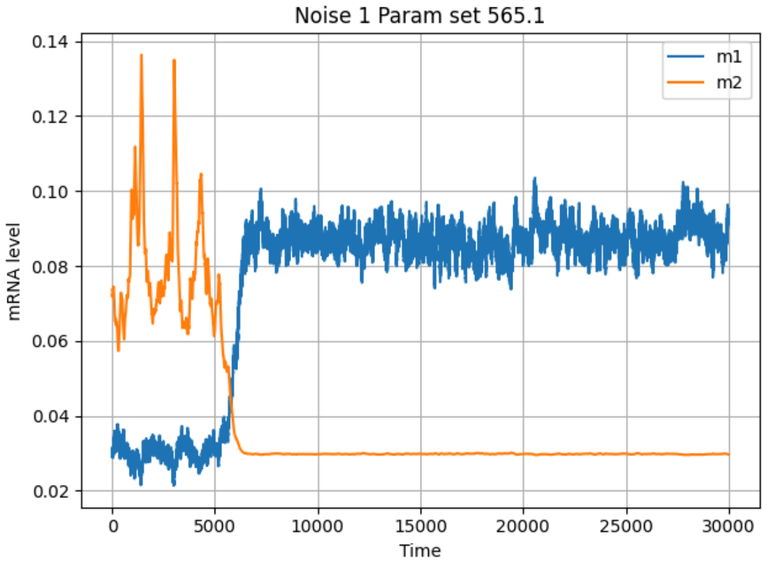
Single switching event

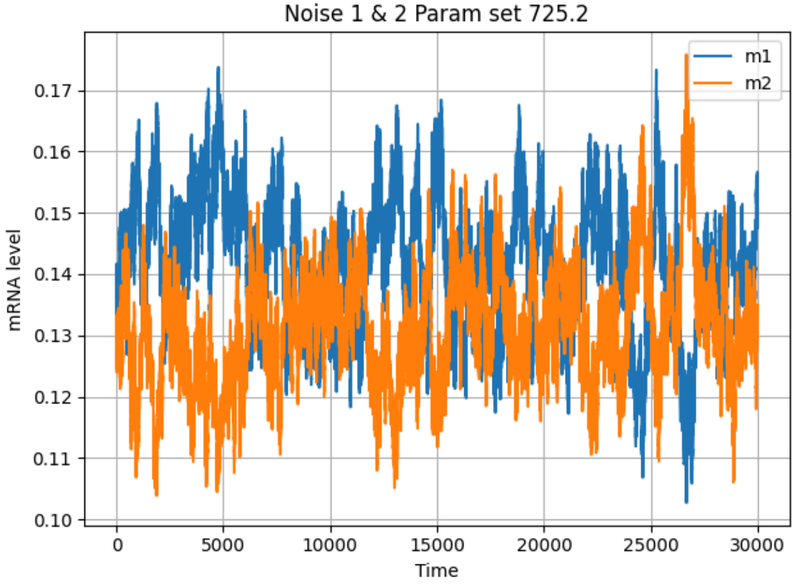


(RARE) spontaneous switching









STATISTICAL METRICS USED FOR ANALYSIS

JENSEN SHANON DISTANCE (JSD)

>> We analyze the expression of stochastic node (node where noise is induced)
>> Comparison of distributions from one parameter set from different initial conditions

$$\mathrm{JSD}(P \parallel Q) = \frac{1}{2} D(P \parallel M) + \frac{1}{2} D(Q \parallel M)$$

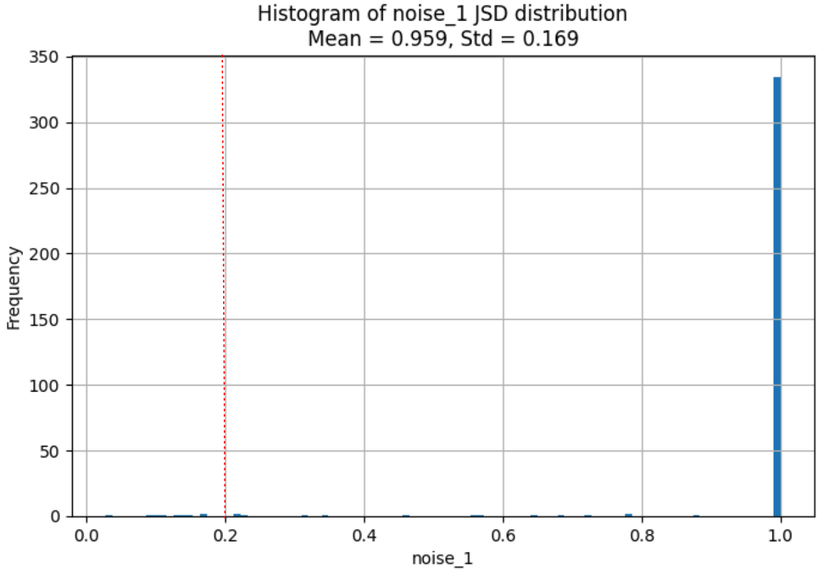
where $M = \frac{1}{2}(P + Q)$ is a mixture distribution of P and Q

Why JSD??
>> Symmetric: JSD(P,Q) == JSD(Q,P)
>> Finite: JSD ϵ [0,1]
>> Numerical stability

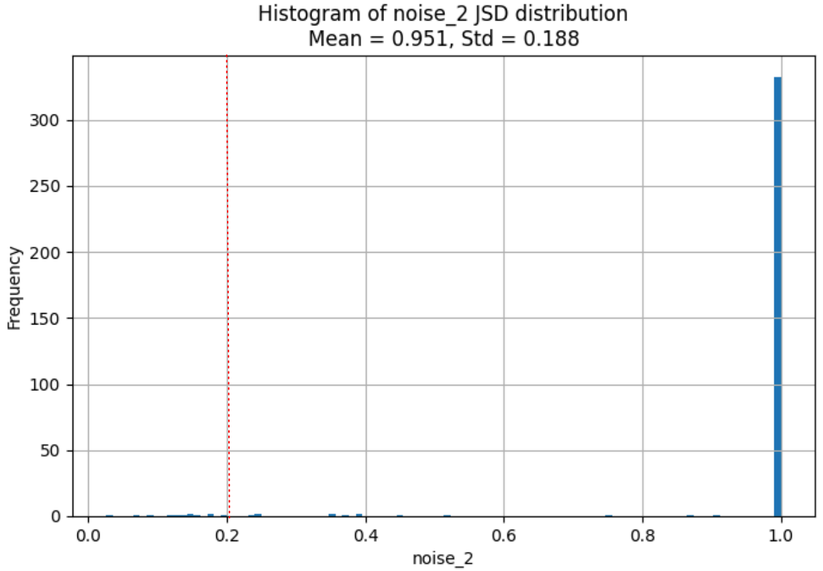
PARTICULAR NOISE INTENSITY → **ALL PARAMETER SETS** → **COMPARE DISTRIBUTIONS FROM INIT. COND. 1 & INIT. COND. 2** → **FILTER PARAMETER SETS WITH LOW JSD** → **RELATIVE EXPRESSION**

>> **LOW JSD** – means – distributions **overlap strongly** with minimal divergence
>> HIGH JSD – means – distributions are significantly distinct

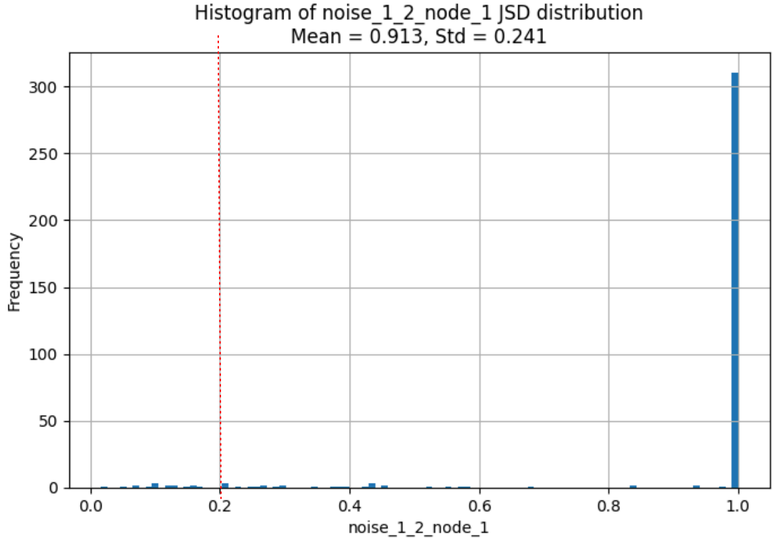
SINGLE NODE NOISE

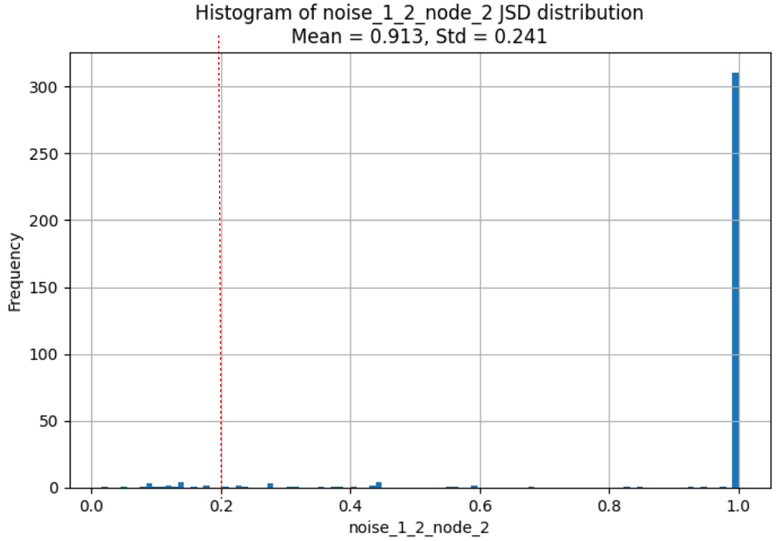


Highly Left skewed



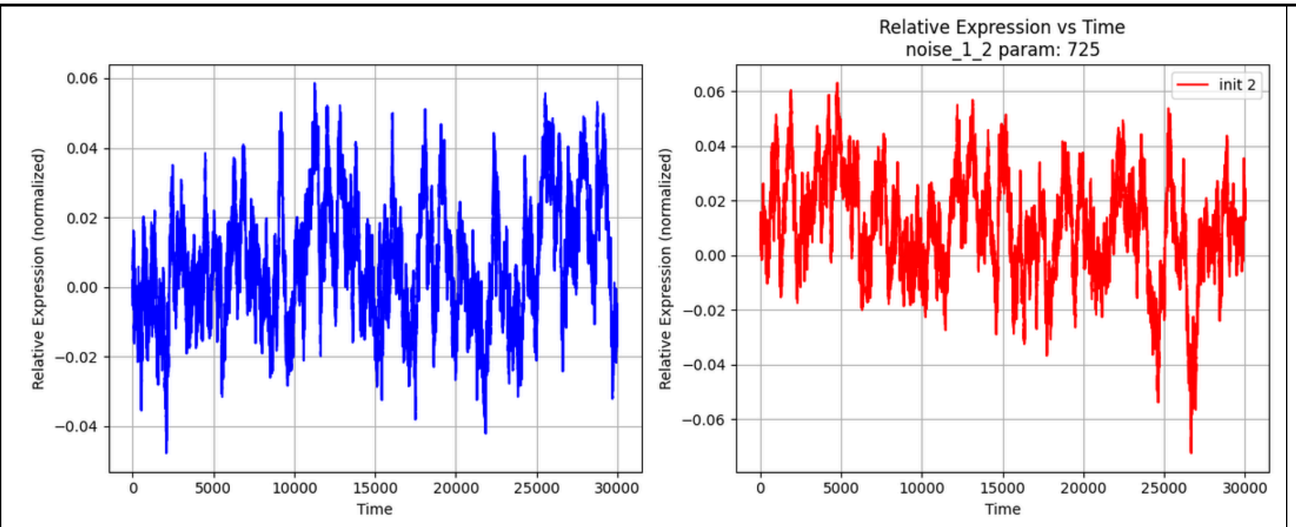
DOUBLE NODE NOISE

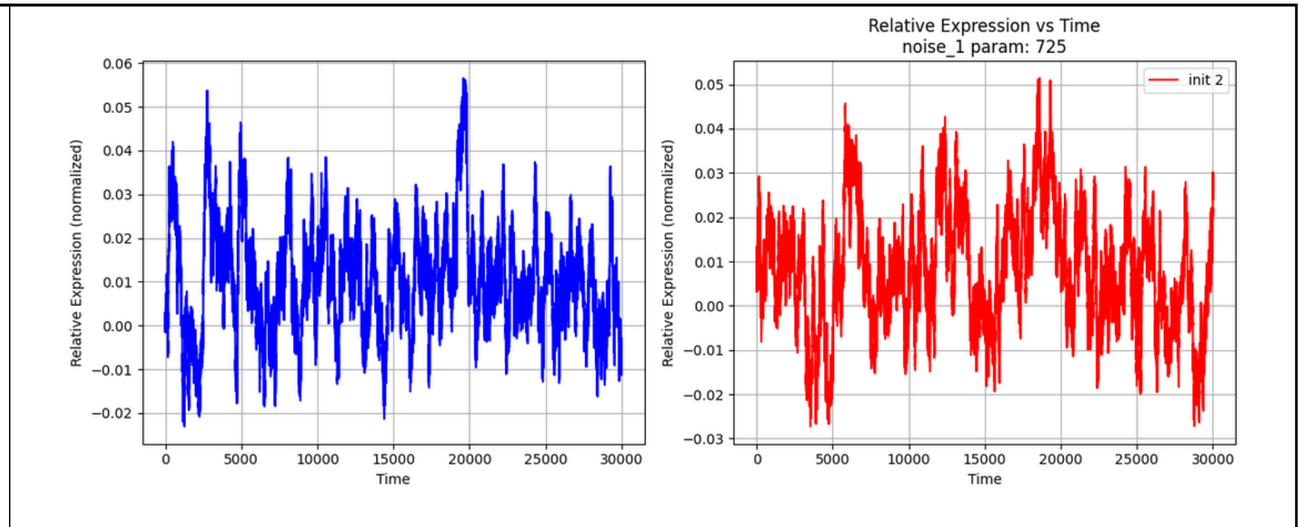


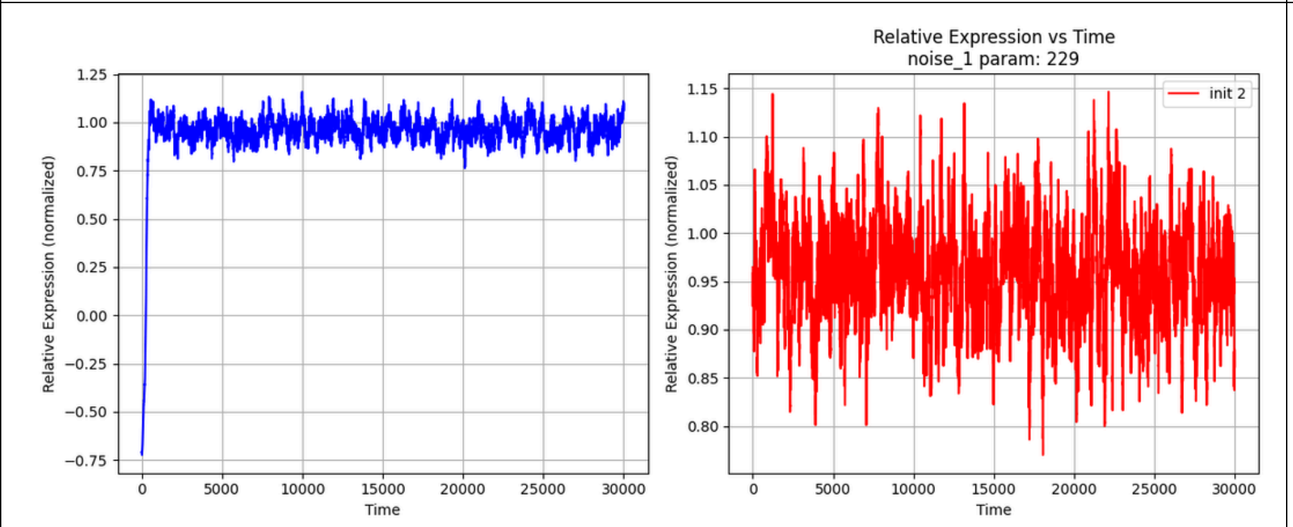


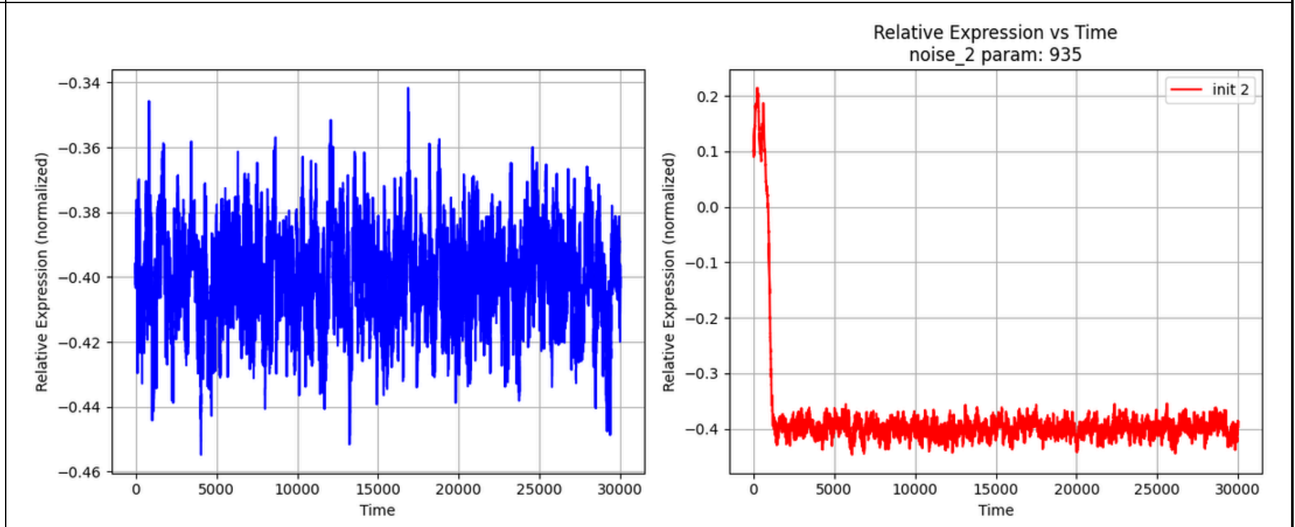
RELATIVE EXPRESSION OF NODE 1 W.R.T NODE 2

TIME SERIES



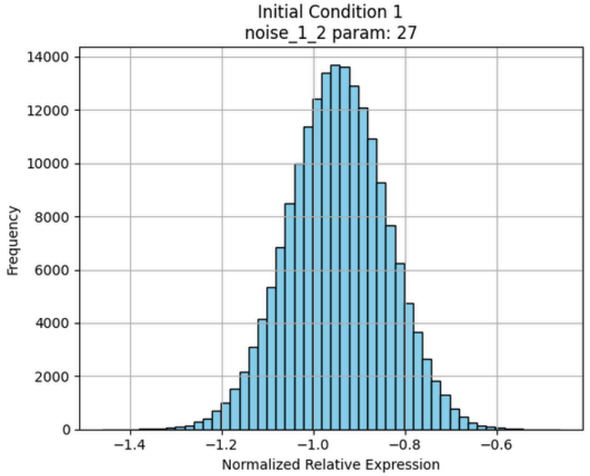


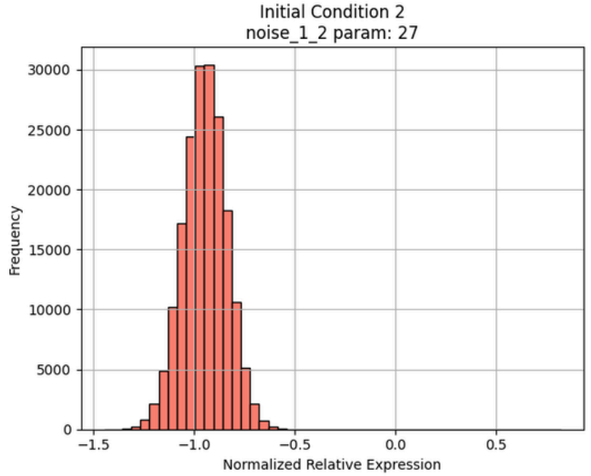


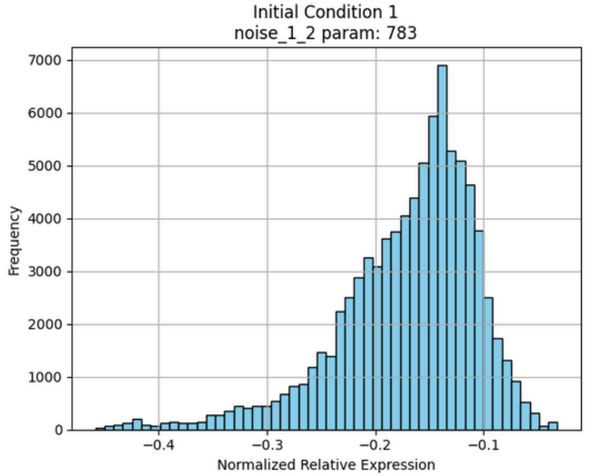


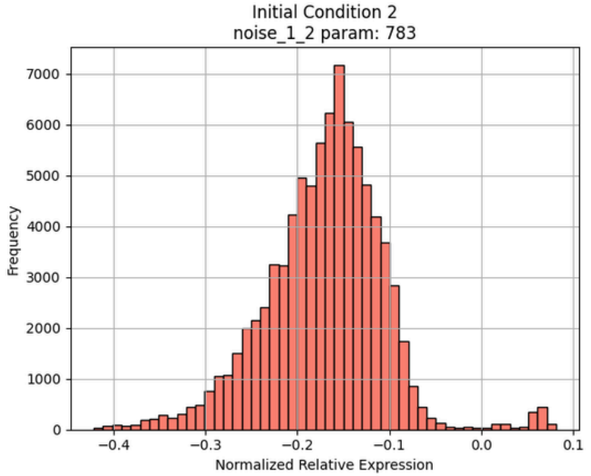
FREQUENCY DISTRIBUTION

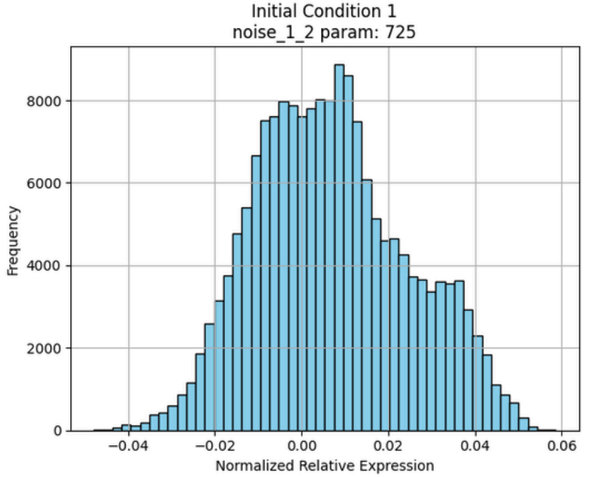
Unimodal distribution

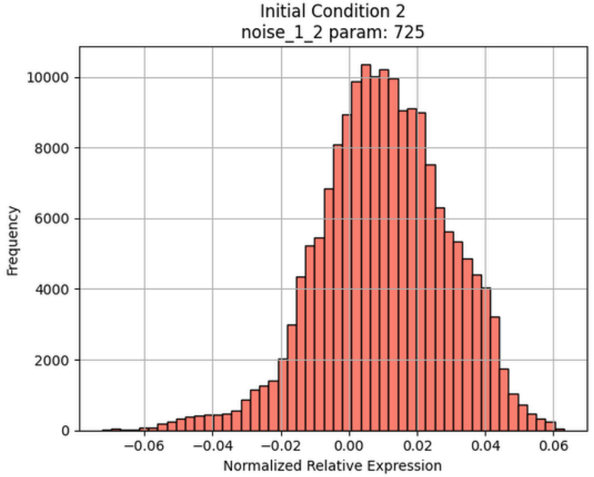


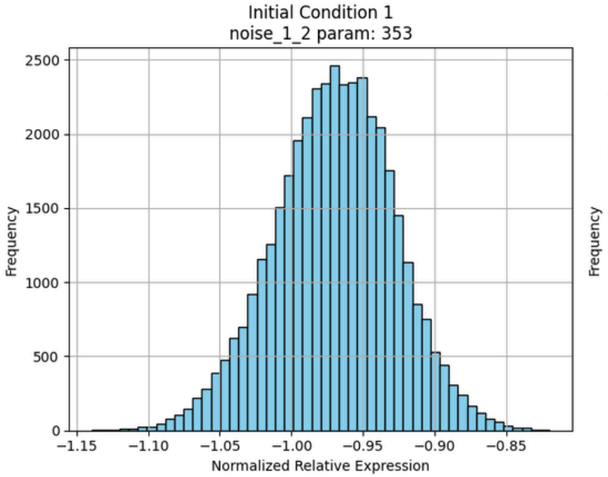


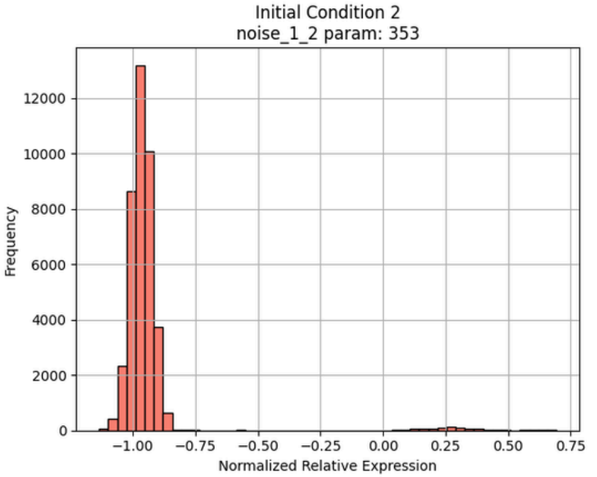




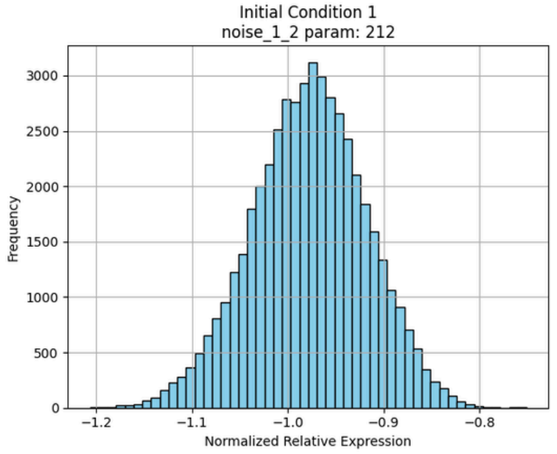


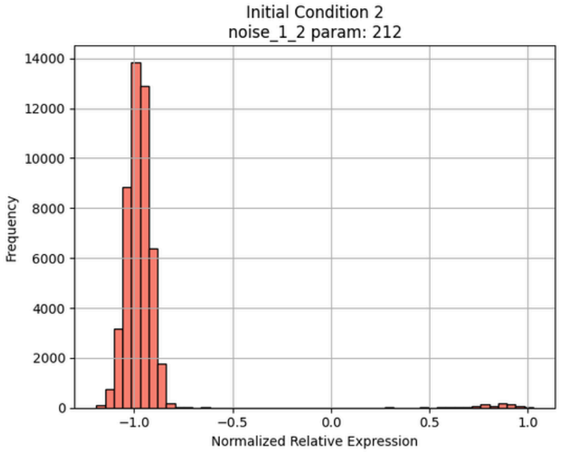


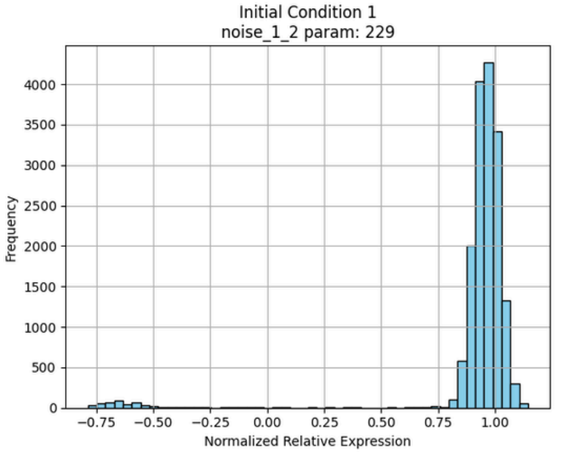


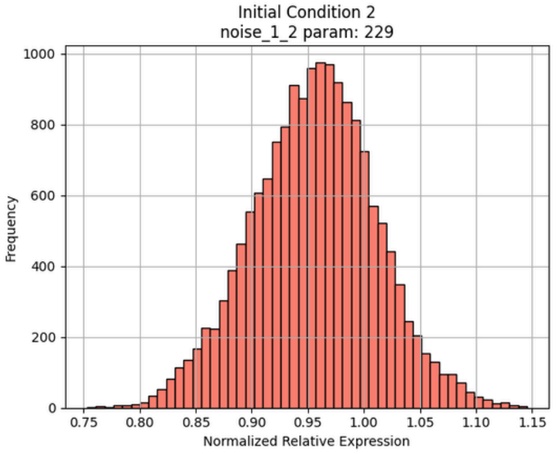


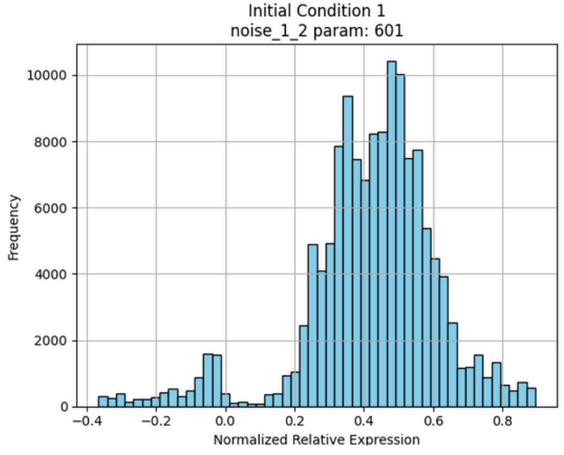
Bimodal distribution

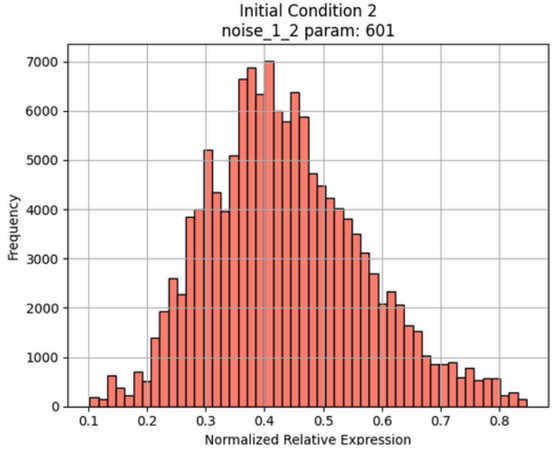


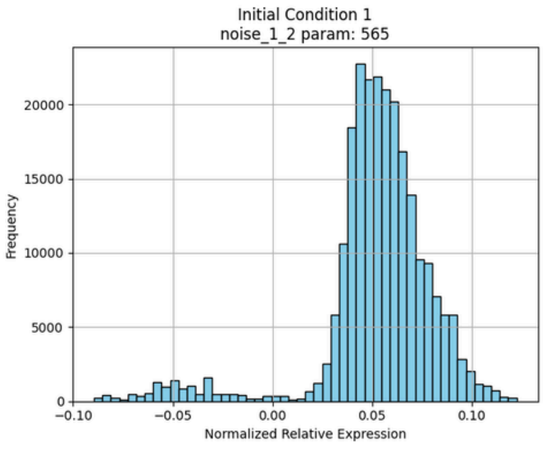


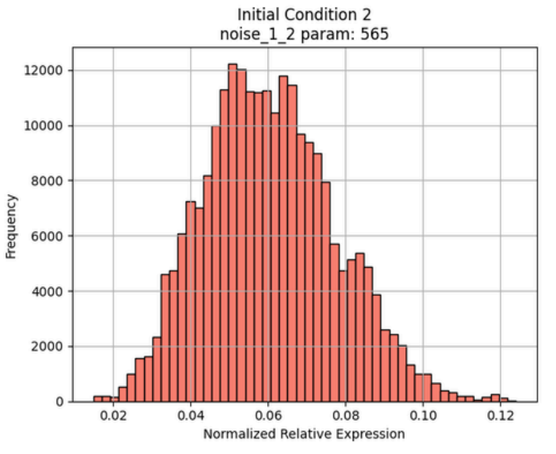


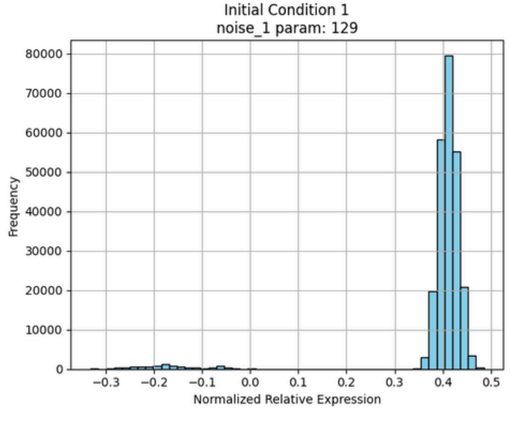


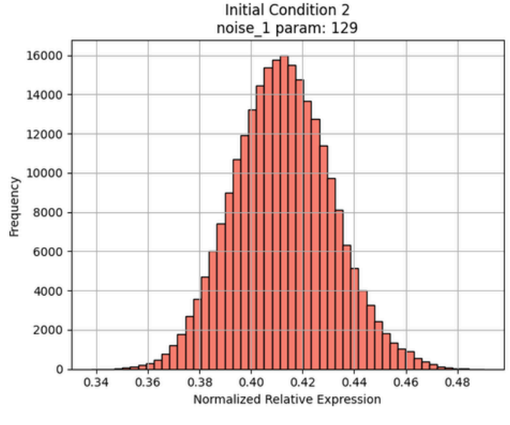


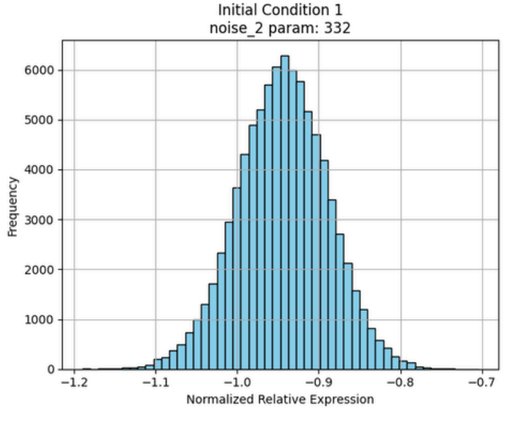


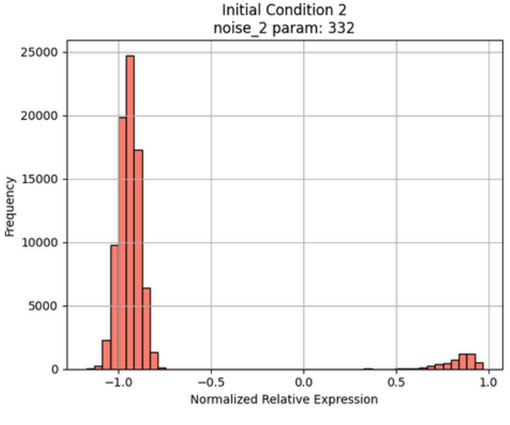






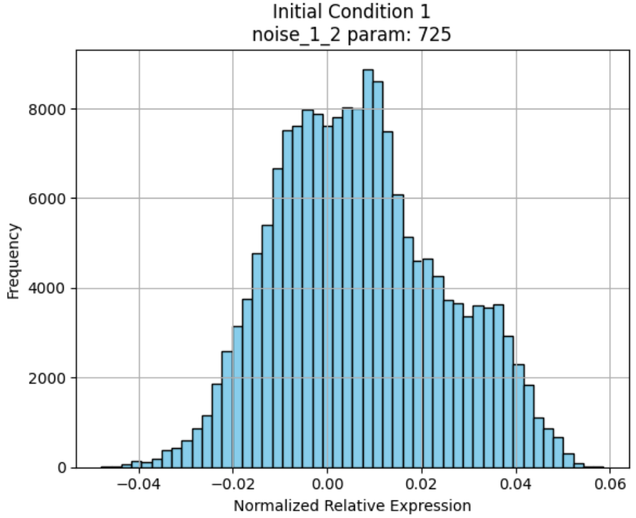


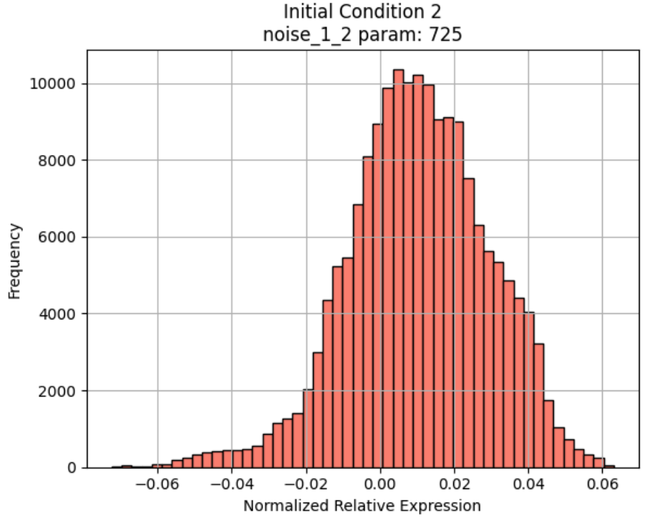


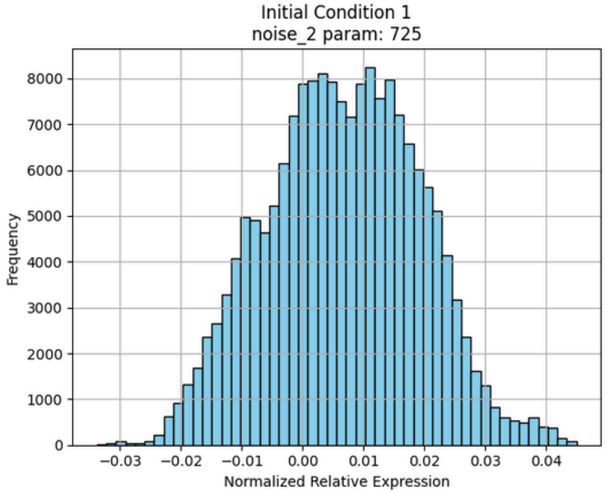


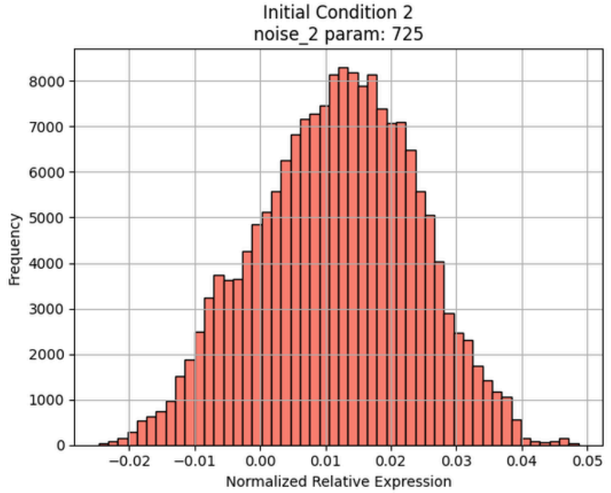
(RARE) Hybrid States

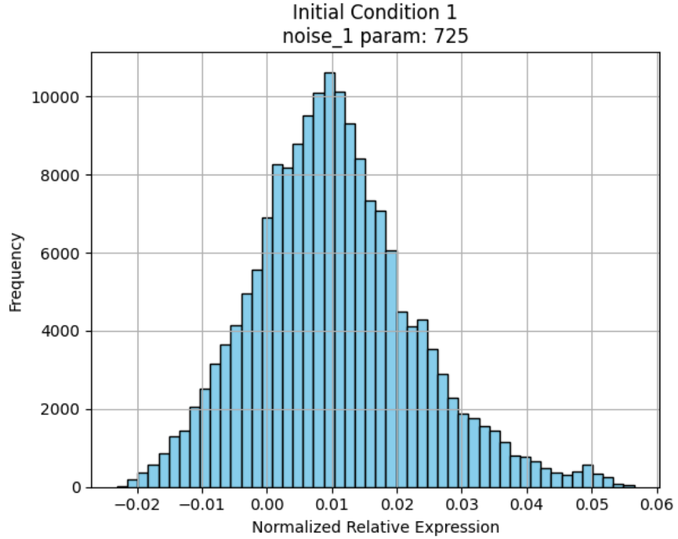
Equal expression of both nodes

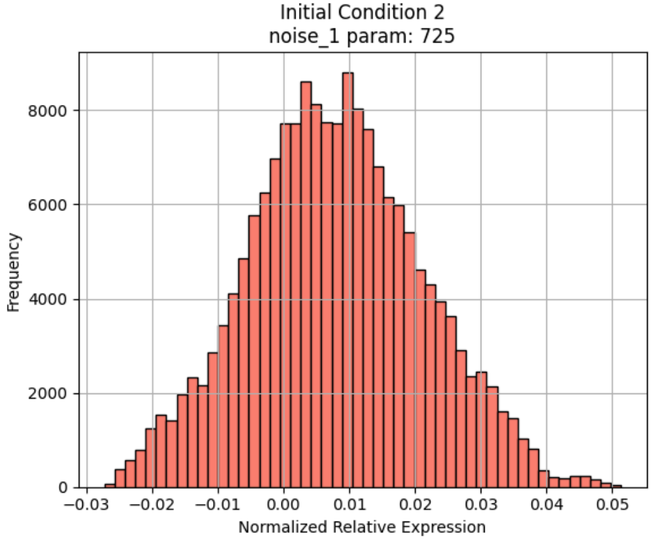












THANK YOU!


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(Supervisor)
Govind
(UG, IISER Pune)
Me
KEEP : GOING